

Reproducing a Depression-Severity Classifier

Reproduction of OPI@LT-EDI-ACL2022: Detecting Signs of Depression from Social Media Text using RoBERTa Pre-trained Language Models (Poswiata & Perelkiewicz, 2022)

Kaggle notebook: kaggle.com/code/vidushigupta1/paper-opi-lt-edi-acl2022-detecting-signs-of-depr

1. Task & Setup

The task is to classify a social media post into one of three depression-severity levels: **not depressed**, **moderately depressed**, or **severely depressed**. This was the shared task at LT-EDI-ACL2022, won by the OPI team using a RoBERTa-large model further pre-trained on a large Reddit corpus ("DepRoBERTa") before fine-tuning on the labeled task.

This report compares two models on the same held-out dev set (1,000 posts):

- **Authors' pretrained model** — the original team's published, fully fine-tuned checkpoint, evaluated as-is.
- **Reproduction** — DepRoBERTa fine-tuned from scratch for this report, using the same base checkpoint but an independent training run (3 epochs, batch size 8, learning rate 1e-5, plain PyTorch training loop).

2. Dataset

Split	Total Rows	Not Depressed	Moderate	Severe
Train	6,006	650	3,101	2,255
Dev (used for evaluation)	1,000	90	510	400

Note the class imbalance: "not depressed" is the minority class in both splits, while "moderate" dominates. This matters for interpreting per-class scores below.

3. Results

3.1 Authors' Pretrained Model

Class	Precision	Recall	F1-score	Support
Not depressed	0.54	0.41	0.47	90
Moderate	0.70	0.76	0.73	510
Severe	0.72	0.68	0.70	400
Accuracy			0.70	1000
Macro avg	0.65	0.62	0.63	1000
Weighted avg	0.69	0.70	0.69	1000

3.2 Reproduction (DepRoBERTa, fine-tuned independently)

Class	Precision	Recall	F1-score	Support
Not depressed	0.49	0.53	0.51	90
Moderate	0.63	0.77	0.69	510
Severe	0.68	0.48	0.56	400
Accuracy			0.63	1000
Macro avg	0.60	0.59	0.59	1000
Weighted avg	0.64	0.63	0.62	1000

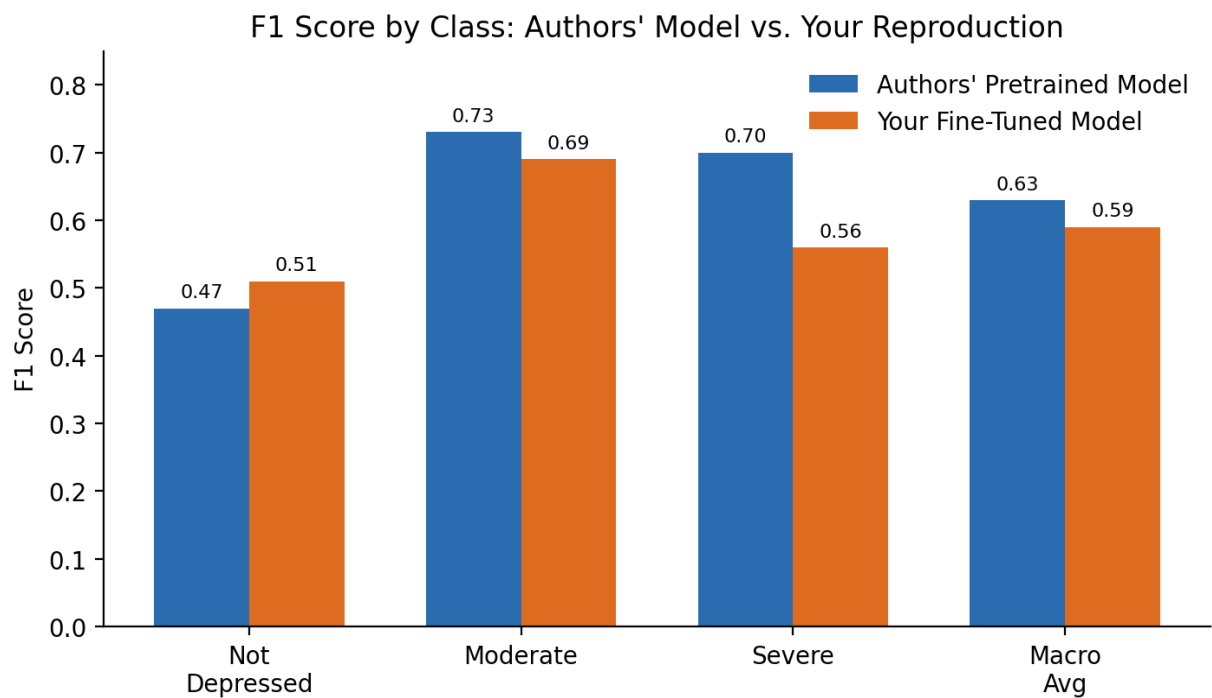


Figure 1. Per-class F1 score, authors' pretrained model vs. independent reproduction. The largest gap appears on the "severe" class.

4. Training Process (Reproduction Only)

The reproduction started from the rafalposwiata/deproberta-large-v1 checkpoint — RoBERTa-large further pre-trained by the original authors on a ~397K-post Reddit depression corpus — and fine-tuned it on the labeled training set for 3 epochs (batch size 8, learning rate 1e-5, linear warmup schedule, AdamW optimizer).

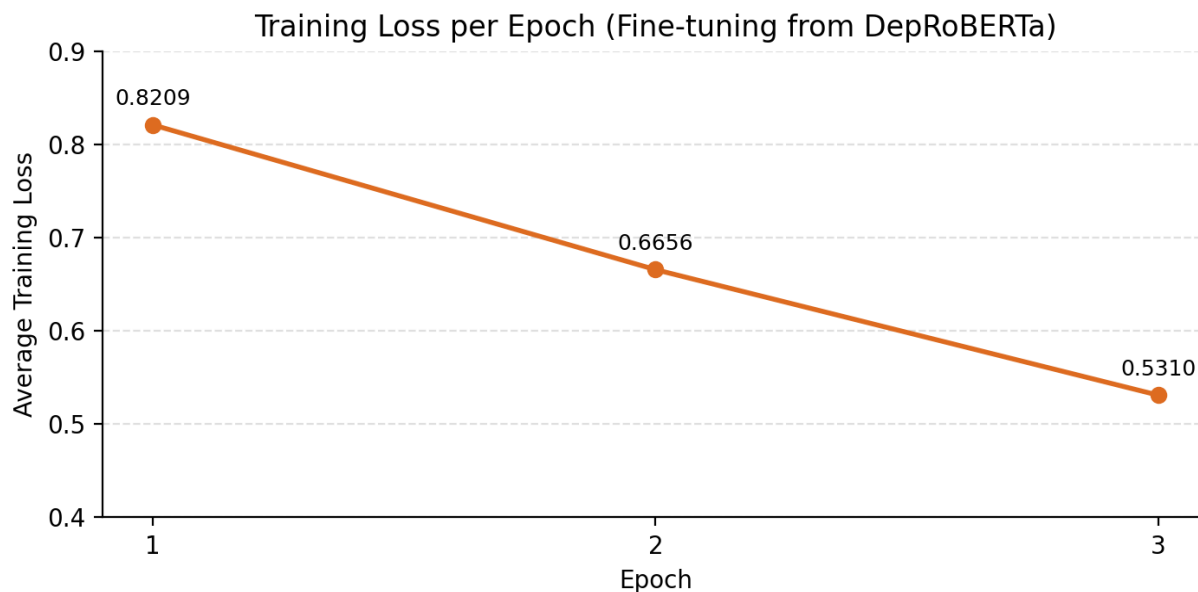


Figure 2. Average training loss per epoch. Loss decreased consistently across all three epochs, indicating the model had not yet converged when training stopped.

5. Head-to-Head Summary

Metric	Authors' Model	Reproduction	Gap
Accuracy	0.70	0.63	-0.07
Macro F1	0.63	0.59	-0.04
Weighted F1	0.69	0.62	-0.07
F1 — Not depressed	0.47	0.51	+0.04
F1 — Moderate	0.73	0.69	-0.04
F1 — Severe	0.70	0.56	-0.14

6. Discussion

The reproduction lands within a reasonable range of the published model but falls short by about 4 points of macro F1 and 7 points of accuracy. This gap is expected rather than a sign of a broken pipeline, for a few concrete reasons:

- **Limited training budget.** The reproduction used a single fixed run of 3 epochs with no dev-set monitoring or early stopping. Training loss was still falling at epoch 3 (0.82 → 0.67 → 0.53), suggesting the model had not converged.
- **No hyperparameter search.** The original published checkpoint likely reflects tuning over learning rate, batch size, and possibly an ensemble — none of which were explored here.
- **Class imbalance hits hardest on "severe."** This class shows the largest single drop (F1 0.70 → 0.56, a 14-point gap), and recall in particular fell sharply (0.68 → 0.48). "Severe" has fewer training examples than "moderate" (2,255 vs 3,101), so a less-tuned model is more likely to default toward the majority class.
- **"Not depressed" reproduced slightly better** (F1 0.51 vs 0.47) — a useful reminder that the gap is not uniform across classes and is worth investigating class-by-class rather than from the headline number alone.

7. Suggested Next Steps

- Add per-epoch dev-set evaluation and early stopping to avoid guessing the right number of epochs.
- Train for more epochs given loss was still decreasing, or raise the learning rate slightly (e.g. 2e-5).
- Apply class-weighted loss to address the "severe" class recall gap specifically.
- Compare against a DeBERTa-v3 fine-tune as a further extension beyond the original paper's architecture.

Note: this is a research reproduction exercise evaluated on a published academic benchmark. It is not a validated diagnostic tool and is not intended for clinical or real-world use on individuals.

Source paper: Poswiata & Perelkiewicz, "OPI@LT-EDI-ACL2022: Detecting Signs of Depression from Social Media Text using RoBERTa Pre-trained Language Models," Proceedings of the Second Workshop on Language Technology for Equality, Diversity and Inclusion, ACL 2022.

Full reproduction code: kaggle.com/code/vidushigupta1/paper-opi-lt-edi-acl2022-detecting-signs-of-depr